

Sales Forecasting & Demand Intelligence System

Executive Report

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Date: July 2026

Executive Summary

This project develops an end-to-end **Sales Forecasting & Demand Intelligence System** that combines machine learning and business analytics to improve retail decision-making. The solution forecasts future sales demand, detects unusual sales behavior, segments customers, and provides an interactive Streamlit dashboard for business users. The objective is to reduce inventory costs, improve demand planning, and support data-driven business decisions.

Business Problem

Retail organizations often face inventory imbalance due to inaccurate demand forecasting. Overstocking increases storage costs, while stock shortages reduce customer satisfaction and revenue. Manual analysis also makes it difficult to identify abnormal sales patterns and high-value customer groups. This project addresses these challenges using predictive analytics and business intelligence.

Dataset & Data Preparation

Dataset: Historical Superstore Retail Sales

Features: 21

Major preprocessing steps

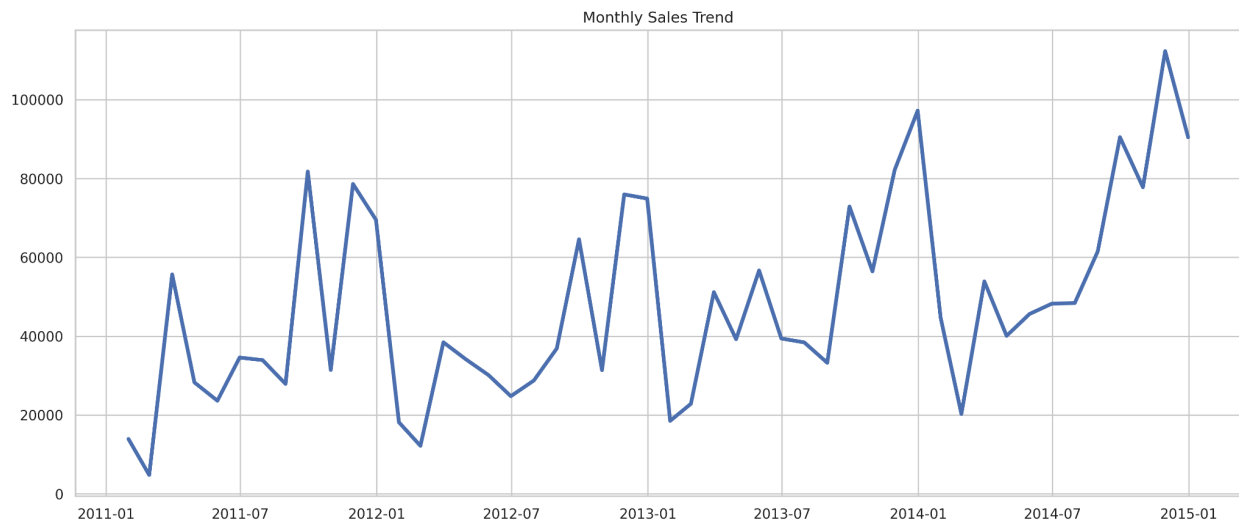
- Removed duplicate records

- Handled missing values
 - Standardized date formats
 - Created new time-based features
 - Generated daily, weekly, monthly, quarterly, and yearly aggregated datasets
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Key Business Insights

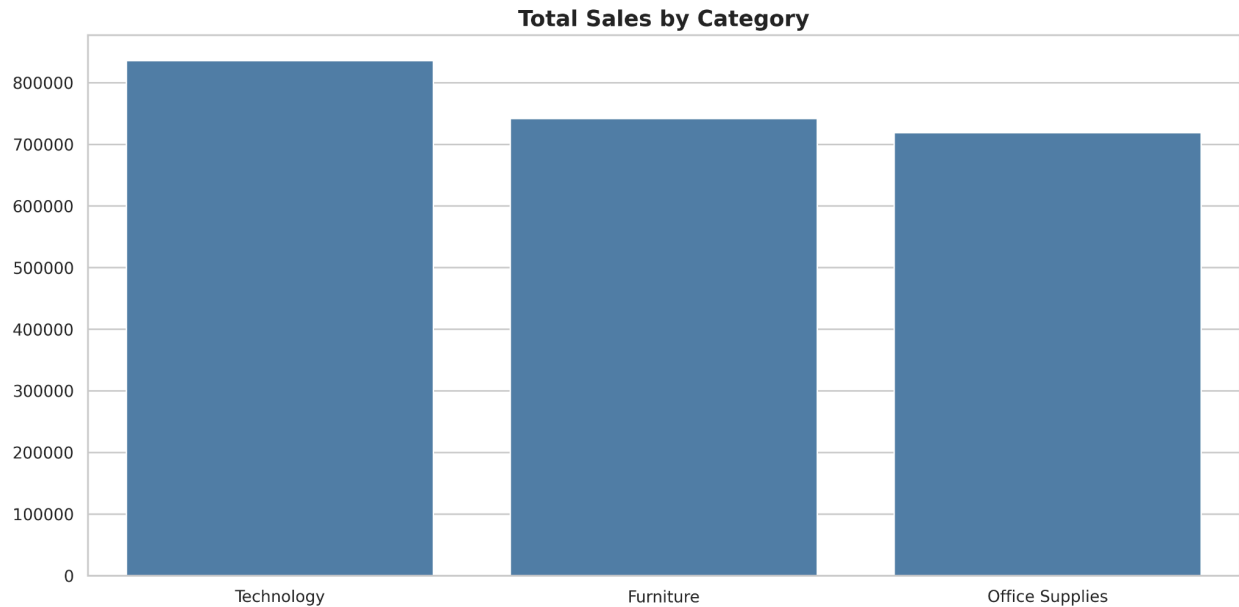
- West and East regions generated the highest sales.
 - Technology and Office Supplies were the strongest performing categories.
 - Furniture showed comparatively lower profitability because of higher discounts.
 - Strong seasonal demand was observed during the final quarter of each year.
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 **Figure 1**



Monthly sales trend highlighting seasonal demand patterns.

 **Figure 2**



Sales distribution across product categories.

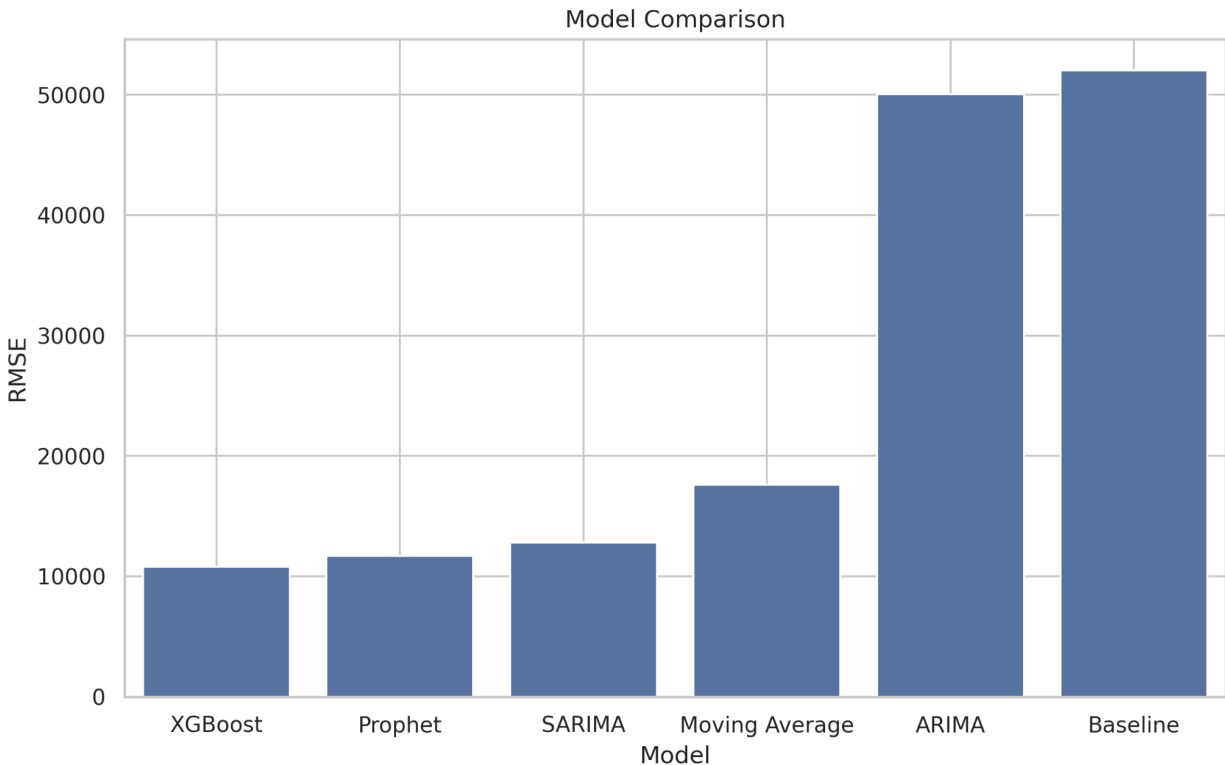
Forecasting & Advanced Analytics

Multiple forecasting approaches were evaluated, including **ARIMA**, **SARIMA**, **Prophet**, and **XGBoost**. Models were compared using **MAE**, **RMSE**, and **MAPE**, and the best-performing model was selected for business forecasting.

The forecasting results provide valuable support for:

- Inventory planning
 - Procurement scheduling
 - Demand forecasting
 - Financial planning
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 **Figure 3**



Performance comparison of forecasting models.

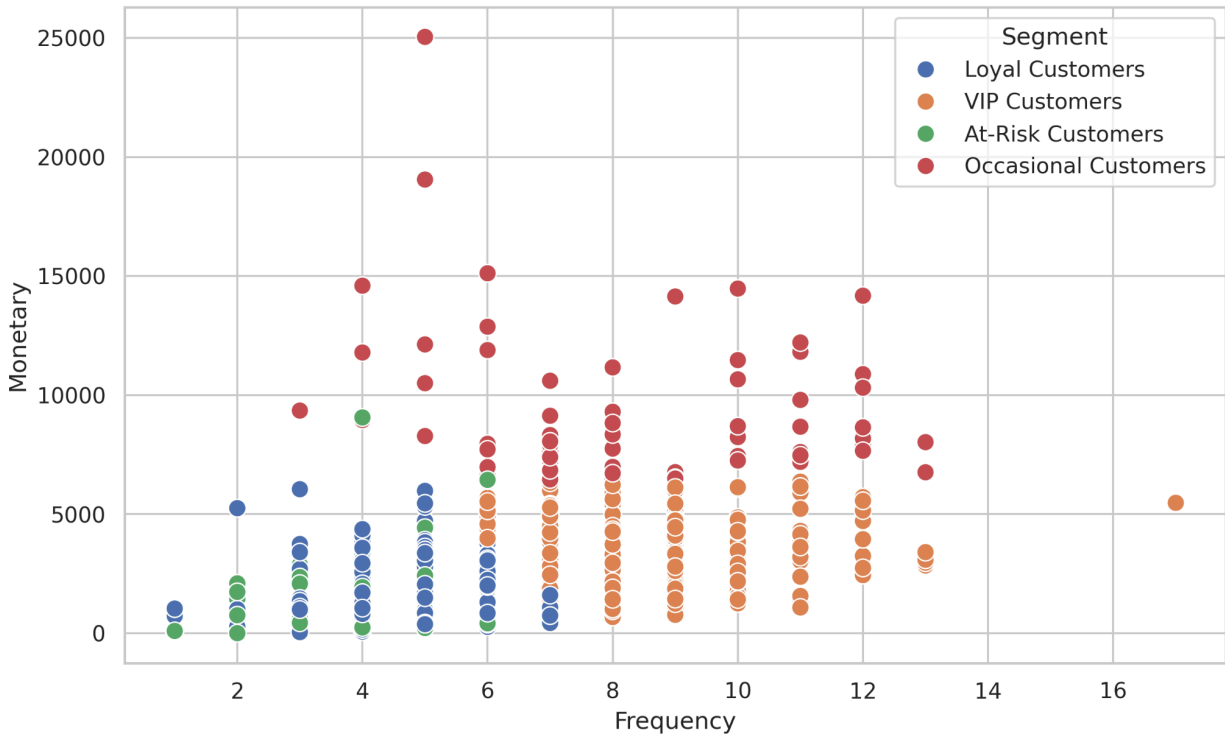
Anomaly Detection

Machine learning techniques identified unusual sales events that may represent promotional campaigns, operational issues, or unexpected market behavior. These insights help businesses investigate abnormal transactions and improve forecasting reliability.

Customer Segmentation

Customers were segmented according to purchasing behavior into groups such as **VIP**, **Loyal**, **Occasional**, and **At-Risk** customers. These segments support personalized marketing strategies and customer retention initiatives.

 **Figure 4**



Customer segments identified using machine learning.

Business Recommendations

- Implement forecasting-driven inventory planning.
 - Optimize discount strategies to improve profitability.
 - Prioritize high-value customer segments through loyalty programs.
 - Monitor detected anomalies to improve operational planning.
 - Use predictive insights for procurement and warehouse management.
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Project Limitation

Forecast accuracy depends on historical sales data and does not account for unexpected external events such as economic disruptions or sudden market changes.

Conclusion

The **Sales Forecasting & Demand Intelligence System** successfully integrates data engineering, forecasting, anomaly detection, customer segmentation, and interactive visualization into a unified business intelligence platform. The solution enables organizations to make informed, data-driven decisions that improve operational efficiency and business performance.

Project Resources

GitHub Repository

<https://github.com/ACT2039/sales-forecasting-demand-intelligence-system>

Streamlit Dashboard

<https://sales-forecasting-demand-intelligence-system-bzf9pd2qmwleoxjp3.streamlit.app/>